EN.553.733 Nonparametric Bayesian Statistics

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Chapter 1

Introduction

The Bayesian point of view with y data and fixed θ paramaters:

$$\mathbb{P}(\theta|y) \propto \mathbb{P}(y|\theta)\mathbb{P}(\theta)$$

Nonparametric Bayes (BNP) extends this by allowing unbounded/growing/infinite parameters

1.1 Exchangeability and De Finetti's Theorem

A theoretical motivation for nonparametric Bayesian methods comes from *De Finetti's Theorem*.

Exchangeability

A sequence $(X_n)_{n\geq 1}$ is *infinitely exchangeable* if for every N and every permutation σ of $\{1,\ldots,N\}$,

$$p(X_1,\ldots,X_N) = p\Big(X_{\sigma(1)},\ldots,X_{\sigma(N)}\Big).$$

This means the joint distribution depends only on the multiset of observations, not their order.

De Finetti Representation

Theorem 1.1.1 (De Finetti (informal))**.** *A sequence $X_1, X_2, \ldots$ is infinitely exchangeable if and only if, for all N, there exists a probability measure P such that*

$$p(X_1,\ldots,X_N) = \int_\Theta \prod_{n=1}^{N} p(X_n \mid \theta)\, P(d\theta).$$

Conditionally on θ, the observations are i.i.d., i.e.,

$$X_n \mid \theta \overset{i.i.d.}{\sim} p(\cdot \mid \theta).$$

Thus exchangeability implies a mixture-of-i.i.d. representation.

Implications

This representation motivates:

- likelihood models $p(x \mid \theta)$,
- prior distributions $P(d\theta)$,
- *nonparametric Bayesian* priors, where P is infinite-dimensional.

1.2 The Bayesian Toolkit

1.2.1 Likelihood

At the core of Bayesian inference is the *likelihood.* Given data $D = x = (x_1, \ldots, x_n)$,

$$L(\theta \mid D) = f(x \mid \theta).$$

If $x_1, \ldots, x_n$ are conditionally independent given θ, then

$$f(x \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

1.2.2 Prior Distribution

Uncertainty about θ is encoded through a *prior* $\pi(\theta)$.

1.2.3 Joint, Marginal, and Posterior

Joint distribution

$$\psi(\theta, x) = f(x \mid \theta)\pi(\theta).$$

Marginal distribution

$$m(x) = \int f(x \mid \theta)\pi(\theta)\, d\theta.$$

This quantity is also called the *evidence* or *model likelihood.*

Posterior distribution

$$\pi(\theta \mid x) = \frac{f(x \mid \theta)\pi(\theta)}{\int f(x \mid \theta)\pi(\theta)\, d\theta} = \frac{f(x \mid \theta)\pi(\theta)}{m(x)}.$$

1.2.4 Predictive Distribution

If $y \sim g(y \mid \theta, x)$, then the predictive distribution is

$$g(y \mid x) = \int g(y \mid \theta, x)\, \pi(\theta \mid x)\, d\theta.$$

This integrates parameter uncertainty into predictions.

Remark While inference focuses on $\pi(\theta \mid x)$, prediction via $g(y \mid x)$ is often the primary goal.

1.3 Conjugate Priors

1.3.1 Definition

Theorem 1.3.1 (Conjugate Prior)**.** *A class $\mathcal{P}$ of prior distributions is conjugate for a likelihood $p(x \mid \theta)$ if*

$$\pi(\theta) \in \mathcal{P} \;\Rightarrow\; \pi(\theta \mid x) \in \mathcal{P}.$$

Examples

- If $x \mid \theta \sim \mathrm{Bin}(n, \theta)$ and $\theta \sim \mathrm{Beta}(a, b)$, then

$$\theta \mid x \sim \mathrm{Beta}(a + x,\ b + n - x).$$

- If $x_1, \ldots, x_n \overset{i.i.d.}{\sim} \mathrm{Poisson}(\lambda)$ and $\lambda \sim \mathrm{Gamma}(a, b)$, then

$$\lambda \mid x \sim \mathrm{Gamma}\Big(a + \textstyle\sum_{i=1}^{n} x_i,\ b + n\Big).$$

1.3.2 Properties

Advantage *Closed-form* posterior updates.

Disadvantage The conjugate family may not reflect prior beliefs.

1.4 Credible Intervals

Theorem 1.4.1 (Credible Interval)**.** *An interval (a, b) is a $(1 - \alpha)100\%$ credible interval for θ if*

$$\Pr(a < \theta < b \mid x) = 1 - \alpha.$$

Quantile Construction

Choose a, b such that

$$\Pr(\theta < a \mid x) = \frac{\alpha}{2}, \qquad \Pr(\theta > b \mid x) = \frac{\alpha}{2}.$$

Remark

Credible intervals quantify posterior uncertainty, unlike frequentist confidence intervals which are defined via repeated sampling.

1.5 MAP Estimator

The *maximum a posteriori (MAP)* estimator is defined as

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} f(x \mid \theta)\pi(\theta).$$

Equivalently,

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \left[\log f(x \mid \theta) + \log \pi(\theta)\right].$$

Motivation

- Interpretable as a *penalized likelihood* estimator.
- Useful when optimization is easier than integration.

1.6 Testing Hypotheses

Hypothesis testing on θ is formulated as

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \notin \Theta_0.$$

1.7 Bayes Factor

Bayesian testing procedures can be based on the posterior probability

$$P^{\pi}(\theta \in \Theta_0 \mid x),$$

or alternatively on the *Bayes factor*.

Definition

The Bayes factor comparing Θ_1 to Θ_0 is

$$B_{10}^{\pi} = \frac{P^{\pi}(\theta \in \Theta_1 \mid x)/P^{\pi}(\theta \in \Theta_0 \mid x)}{P^{\pi}(\theta \in \Theta_1)/P^{\pi}(\theta \in \Theta_0)}.$$

Model Comparison Formulation

For corresponding models $\mathcal{M}_1$ versus $\mathcal{M}_0$,

$$B_{10}^{\pi} = \frac{P^{\pi}(\mathcal{M}_1 \mid x)}{P^{\pi}(\mathcal{M}_0 \mid x)} \bigg/ \frac{P^{\pi}(\mathcal{M}_1)}{P^{\pi}(\mathcal{M}_0)}.$$

If the prior is written as

$$\pi(\theta) = \Pr(\theta \in \Theta_1)\, \pi_1(\theta) + \Pr(\theta \in \Theta_0)\, \pi_0(\theta),$$

then

$$B_{10}^{\pi} = \frac{\int f(x \mid \theta_1)\pi_1(\theta_1)\, d\theta_1}{\int f(x \mid \theta_0)\pi_0(\theta_0)\, d\theta_0} = \frac{m_1(x)}{m_0(x)}.$$

Thus the Bayes factor is *akin to a likelihood ratio.*

1.8 Jeffreys' Scale

Interpretation of the Bayes factor is often based on $\log_{10}(B_{10}^{\pi})$:

- if $\log_{10}(B_{10}^{\pi}) \in [0, 0.5]$, evidence against H_0 is poor,
- if $\log_{10}(B_{10}^{\pi}) \in [0.5, 1]$, it is substantial,
- if $\log_{10}(B_{10}^{\pi}) \in [1, 2]$, it is strong,
- if $\log_{10}(B_{10}^{\pi}) > 2$, it is decisive.

1.9 Markov Chain Monte Carlo (MCMC)

Statistical inference is based on the posterior distribution

$$\pi(\theta \mid x) = \frac{\pi(\theta) f(x \mid \theta)}{\int \pi(\theta) f(x \mid \theta)\, d\theta}.$$

Motivation

- In conjugate cases (e.g. normal-normal, binomial-beta), integrals can be computed analytically.
- In general, closed-form expressions for $\pi(\theta \mid x)$ are not available.
- We still need to compute:
  - point estimates,
  - interval estimates,
  - hypothesis tests.

This motivates simulation-based methods.

The MCMC Principle

Definition

An *MCMC method* for simulating from a distribution π is any method that constructs an ergodic Markov chain (X_n) whose stationary distribution is π.

Given an initial value $x^{(0)}$, the chain is generated via a transition kernel such that

$$X_n \xrightarrow{d} X \sim \pi.$$

Construction

To build such a chain, common methods include:

- Metropolis–Hastings,

- Gibbs sampling.

There are also diagnostic methods to assess whether the chain has reached its stationary distribution.

Metropolis–Hastings Algorithm

Algorithm

Given current state $x^{(t)}$:

- Generate proposal $Y_t \sim q(y \mid x^{(t)})$ (*proposal distribution*).
- Set
$$X^{(t+1)} = \begin{cases} Y_t, & \text{with probability } \rho(x^{(t)}, Y_t), \\ x^{(t)}, & \text{with probability } 1 - \rho(x^{(t)}, Y_t), \end{cases}$$
where
$$\rho(x, y) = \min\left\{ \frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)},\ 1 \right\}.$$

Remarks on Metropolis–Hastings

- The quantity $\rho(x, y)$ is the *acceptance ratio*.
- The resulting chain is Markov since $X^{(t+1)}$ depends only on $X^{(t)}$.
- If $q(x \mid y) = q(y \mid x)$, the proposal is symmetric and the method reduces to the *Metropolis algorithm*.
- In this case, the acceptance probability simplifies to
$$\frac{\pi(y_t)}{\pi(x^{(t)})}.$$

1.10 Additional Remarks

Acceptance Mechanism

If
$$\frac{\pi(y_t)}{q(y_t \mid x^{(t)})} > \frac{\pi(x_t)}{q(x^{(t)} \mid y_t)},$$
the proposal is always accepted. Otherwise, it is accepted with probability ρ.

Invariance

The algorithm depends only on ratios:
$$\frac{\pi(y_t)}{\pi(x^{(t)})}, \qquad \frac{q(x^{(t)} \mid y_t)}{q(y_t \mid x^{(t)})}.$$

Thus, normalizing constants cancel out, provided they do not depend on the conditioning variables.

Choice of Proposal

The choice of proposal distribution q strongly affects performance (efficiency and convergence).

A good proposal should:

- be easy to sample from,
- allow easy computation of the acceptance ratio,
- move a reasonable distance in parameter space,
- not be rejected too frequently.

1.11 Gibbs Sampling

1.11.1 Overview

The Gibbs sampler is one of the most widely used computational methods for Bayesian inference. It was first used in statistical physics and later popularized in statistics.

The method is particularly useful for high-dimensional target distributions.

1.11.2 Key Idea

The Gibbs sampler works by *sequentially sampling from univariate conditional distributions* rather than attempting to sample directly from a multivariate distribution.

1.11.3 Two-Stage Gibbs Sampler

Consider a bivariate posterior distribution

$$\pi(\theta, \lambda \mid x).$$

Algorithm

- **Initialization:** Start with an arbitrary value $\lambda^{(0)}$.
- **Iteration t:**
  - Sample
    $$\theta^{(t)} \sim \pi_1(\theta \mid x, \lambda^{(t-1)}),$$
  - Sample
    $$\lambda^{(t)} \sim \pi_2(\lambda \mid x, \theta^{(t)}).$$

The joint distribution $\pi(\theta, \lambda \mid x)$ is a stationary distribution for this Markov chain.

1.11.4 Multi-Stage Gibbs Sampler

General Form

For a joint distribution $\pi(\theta \mid x)$ with full conditionals $\pi_1, \dots, \pi_p$, define the iterative scheme:
Given $(\theta_1^{(t)}, \dots, \theta_p^{(t)})$,

$$\begin{aligned}
\theta_1^{(t+1)} &\sim \pi_1(\theta_1 \mid x, \theta_2^{(t)}, \dots, \theta_p^{(t)}), \\
\theta_2^{(t+1)} &\sim \pi_2(\theta_2 \mid x, \theta_1^{(t+1)}, \theta_3^{(t)}, \dots, \theta_p^{(t)}), \\
&\vdots \\
\theta_p^{(t+1)} &\sim \pi_p(\theta_p \mid x, \theta_1^{(t+1)}, \dots, \theta_{p-1}^{(t+1)}).
\end{aligned}$$

Then

$$\theta^{(t)} \;\to\; \theta \sim \pi(\theta \mid x).$$

1.11.5 Remarks on Gibbs Sampling

- The order of updates is not essential and can be changed without affecting validity.
- The update order can also be randomized (*random scan Gibbs*).
- It is not necessary to update every component at each iteration, provided each component is updated sufficiently often.
- The overall process defines a Markov chain, but the individual component sequences $\theta_i^{(t)}$ are generally not Markov.

1.11.6 Convergence Diagnostics

An MCMC algorithm is said to have converged at iteration T if for all $t > T$, the samples can be regarded as drawn from the stationary distribution.

- Lack of convergence leads to biased and unreliable estimates.
- Diagnostics are used to detect signs of non-convergence.
- Passing a diagnostic does *not* guarantee stationarity.

Common Diagnostics

- Geweke
- Gelman and Rubin
- Heidelberger and Welch
- Raftery and Lewis

See the *R package CODA* for implementation.

Quick Checks of Convergence

- A *trace plot* of $\theta^{(t)}$ against t provides a basic visual diagnostic.
- Chains may appear to converge if trapped near a local mode.
- Running multiple chains from different starting values helps assess convergence.

1.12 Autocorrelation

Mixing

The speed at which the chain explores the parameter space is called *mixing*.

High autocorrelation implies slow mixing and slow convergence.

Autocorrelation Function

For a sequence $\{\theta^{(1)}, \dots, \theta^{(T)}\}$, the lag-$t$ autocorrelation function is

$$\mathrm{acf}_t(\theta) = \frac{\frac{1}{T-t}\sum_{j=1}^{T-t}(\theta^{(j)} - \bar{\theta})(\theta^{(j+t)} - \bar{\theta})}{\frac{1}{T-t}\sum_{j=1}^{T}(\theta^{(j)} - \bar{\theta})^2}.$$

Remarks

- Gibbs sampling offers limited control over correlation structure.
- In Metropolis-type algorithms, correlation can be adjusted through the proposal distribution.
- Adaptive variants modify the proposal to improve mixing.

Practical Implications

- High autocorrelation indicates slow convergence.
- Thinning may be used to reduce correlation between retained samples.

1.13 Reading: Mixture Models and Bayesian Computation

Motivation for Mixtures

Mixture models provide a flexible framework for modelling complex data distributions:

$$f(x) = \sum_{i=1}^{k} p_i f(x \mid \theta_i), \quad \sum_{i=1}^{k} p_i = 1.$$

They serve two key roles:

- *Latent structure modelling*: data arise from multiple subpopulations (clustering interpretation)
- *Approximation*: mixtures act as flexible approximations to unknown densities

Thus, mixtures bridge parametric and nonparametric modeling.

Latent Variable Representation

Introduce allocation variables

$$Z_i \sim \text{Multinomial}(1; p_1, \ldots, p_k), \quad X_i \mid Z_i = j \sim f(x \mid \theta_j).$$

This transforms the model into a *complete-data formulation*, which is critical for computation.

Computational Challenge

The likelihood for n observations is

$$L(\theta, p \mid x) = \prod_{i=1}^{n} \sum_{j=1}^{k} p_j f(x_i \mid \theta_j).$$

Expanding this produces k^n terms, making direct inference intractable.
Implication: Closed-form Bayesian inference is generally impossible.

Combinatorial Explosion

Posterior inference can be written as

$$\pi(\theta, p \mid x) = \sum_{z} \omega(z)\, \pi(\theta, p \mid x, z),$$

where z are allocation vectors.

- Number of allocations: k^n
- Only a *tiny subset* have non-negligible weight

This motivates stochastic approximation methods (MCMC).

EM vs Bayesian Computation

The EM algorithm alternates:

- E-step: compute expected allocations
- M-step: maximize expected complete likelihood

While efficient for MLE, EM:

- converges only to *local modes*

- is sensitive to initialization

Key contrast:

- EM: optimization
- MCMC: full posterior exploration

Mixtures as Ill-Posed Problems

Mixture inference is inherently unstable:

- Some components may receive *no data*
- Likelihood can become *unbounded*
- Small data changes $\Rightarrow$ large parameter changes

This is a classic *inverse problem*.

Label Switching and Identifiability

The likelihood is invariant under permutation:

$$(\theta_1, \ldots, \theta_k) \sim (\theta_{\sigma(1)}, \ldots, \theta_{\sigma(k)}).$$

Consequences:

- Posterior has $\mathcal{O}(k!)$ symmetric modes
- Marginal posteriors for components are identical
- Standard estimators (means) are meaningless

This is known as the *label switching problem*.

Prior Design in Mixtures

Improper priors are problematic:

$$\pi(\theta) = \prod_i \pi_i(\theta_i) \quad \Rightarrow \quad \text{posterior may be improper.}$$

Reason:

- Some components may have no observations
- Their posterior remains equal to the prior

Solution:

- Use proper priors
- Introduce dependence between components

Mixtures and Nonparametric Bayes

Mixtures naturally connect to nonparametric Bayesian methods:

- Kernel density estimation:
$$\hat{f}(x) = \frac{1}{n}\sum_{i=1}^{n} K(x - x_i)$$
is a mixture model
- Dirichlet process:
$$F \sim \mathrm{DP}(F_0, \alpha)$$
induces an *infinite mixture representation*
- Bernstein polynomials:
$$f(x) = \sum_k p_k \mathrm{Beta}(\alpha_k, \beta_k)$$
approximate arbitrary densities

Thus, Bayesian nonparametrics can be viewed as *mixtures with infinitely many components*.

Key Takeaways for MCMC

- Mixture models are computationally intractable in closed form
- Latent variables enable tractable conditional structure
- MCMC (Gibbs, MH) exploits this structure
- Posterior geometry is multimodal and symmetric
- Careful diagnostics and interpretation are required

Chapter 2

Dirichlet Process

Bayesian Models Revisited

A Bayesian statistical model consists of:

- a likelihood $f(y \mid \theta)$,
- a prior distribution $\pi(\theta)$.

Uncertainty about parameters θ is modeled through the prior distribution.
Inference is based on the posterior distribution

$$\pi(\theta \mid y) = \frac{f(y \mid \theta)\pi(\theta)}{\int f(y \mid \theta)\pi(\theta)\, d\theta}.$$

2.1 Nonparametric Bayesian Priors

Modeling Idea

A *nonparametric Bayesian model* places a distribution on distributions.

$$\theta \mid G \sim G$$

where G itself is random:

$$G \sim P(G).$$

Thus, uncertainty is modeled at two levels:

- uncertainty in θ,
- uncertainty in the distribution G.

Inference may target both θ and G.

2.2 Finite Mixture Model

Generative Model (Two Components)

Consider a Gaussian mixture model with $K = 2$:

- Latent assignments:
$$z_n \overset{iid}{\sim} \text{Categorical}(\rho_1, \rho_2),$$
- Observations:
$$x_n \mid z_n \sim \mathcal{N}(\mu_{z_n}, \Sigma),$$
- Component means:
$$\mu_k \overset{iid}{\sim} \mathcal{N}(\mu_0, \Sigma_0),$$
- Mixing weights:
$$\rho_1 \sim \text{Beta}(a_1, a_2), \quad \rho_2 = 1 - \rho_1.$$

Inference Goal

- assign data points to clusters,
- estimate cluster parameters.

2.3 Beta Distribution Review

The Beta distribution is

$$\text{Beta}(\rho_1 \mid a_1, a_2) = \frac{\Gamma(a_1 + a_2)}{\Gamma(a_1)\Gamma(a_2)} \rho_1^{a_1 - 1}(1 - \rho_1)^{a_2 - 1}.$$

Gamma Function

- For integer m: $\Gamma(m + 1) = m!$
- For $x > 0$: $\Gamma(x + 1) = x\Gamma(x)$

Behavior

- $a_1 = a_2 \to 0$: mass near boundaries
- $a_1 = a_2 \to \infty$: concentration around 1/2
- $a_1 > a_2$: skew toward 1

Conjugacy

If

$$\rho_1 \sim \mathrm{Beta}(a_1, a_2), \quad z \sim \mathrm{Categorical}(\rho_1, \rho_2),$$

then

$$\rho_1 \mid z \sim \mathrm{Beta}(a_1 + \mathbf{1}\{z = 1\},\ a_2 + \mathbf{1}\{z = 2\}).$$

2.4 Dirichlet Distribution

Definition

For $\rho_{1:K}$,

$$\mathrm{Dirichlet}(\rho_{1:K} \mid a_{1:K}) = \frac{\Gamma\left(\sum_{k=1}^K a_k\right)}{\prod_{k=1}^K \Gamma(a_k)} \prod_{k=1}^K \rho_k^{a_k - 1},$$

with

$$\rho_k > 0, \quad \sum_{k=1}^K \rho_k = 1.$$

Construction via Gamma

Let

$$Z_j \sim \mathrm{Gamma}(a_j, 1), \quad j = 1, \ldots, K,$$

and define

$$Y_j = \frac{Z_j}{\sum_{l=1}^K Z_l}.$$

Then

$$(Y_1, \ldots, Y_K) \sim \mathrm{Dirichlet}(a_1, \ldots, a_K).$$

Special Case

For $K = 2$:

$$\mathrm{Dirichlet}(a_1, a_2) = \mathrm{Beta}(a_1, a_2).$$

Moments

$$E(Y_j) = \frac{a_j}{\sum_{l=1}^K a_l}, \quad E(Y_j^2) = \frac{a_j(a_j + 1)}{(\sum_l a_l)(1 + \sum_l a_l)}.$$

For $i \neq j$:

$$E(Y_i Y_j) = \frac{a_i a_j}{(\sum_l a_l)(1 + \sum_l a_l)}.$$

Marginals

$$Y_j \sim \text{Beta}\left(a_j,\ \sum_{i \neq j} a_i\right).$$

Conjugacy

If

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}), \quad z \sim \text{Categorical}(\rho_{1:K}),$$

then

$$\rho_{1:K} \mid z \sim \text{Dirichlet}(a'_1, \ldots, a'_K), \quad a'_k = a_k + \mathbf{1}\{z = k\}.$$

2.5 Finite Mixture Model (General K)

- Mixing weights:
$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}),$$
- Component means:
$$\mu_k \sim \mathcal{N}(\mu_0, \Sigma_0),$$
- Assignments:
$$z_n \sim \text{Categorical}(\rho_{1:K}),$$
- Observations:
$$x_n \mid z_n \sim \mathcal{N}(\mu_{z_n}, \Sigma).$$

2.6 When $K > N$

Interpretation

- K: number of possible latent groups (components)
- clusters: number of components actually represented in the data

Key Observations

- Number of clusters is random and typically less than K
- Number of clusters grows with sample size N

Motivation

Choosing a finite K is difficult:

- K is unknown in practice
- inference depends heavily on K
- problematic for streaming or growing datasets

2.7 Choosing $K = \infty$

Motivation

In mixture models, choosing a finite K is difficult:

- K is unknown a priori,
- inference depends strongly on K,
- problematic for streaming or growing datasets.

This motivates considering the limit

$$K = \infty.$$

Infinite Mixing Weights

We seek a sequence of strictly positive weights

$$\rho = (\rho_1, \rho_2, \ldots) \quad \text{such that} \quad \sum_{k=1}^{\infty} \rho_k = 1.$$

Dirichlet Decomposition

From the finite Dirichlet:

$$\rho_{1:K} \sim \mathrm{Dirichlet}(a_{1:K}),$$

we have the decomposition:

$$\rho_1 \sim \mathrm{Beta}\Big(a_1,\ \sum_{k=2}^{K} a_k\Big),$$

and conditionally,

$$\frac{(\rho_2, \ldots, \rho_K)}{1 - \rho_1} \sim \mathrm{Dirichlet}(a_2, \ldots, a_K).$$

This suggests a recursive construction.

2.8 Stick-Breaking Construction

2.8.1 Finite Construction

Define:

$$V_1 \sim \mathrm{Beta}(a_1, a_2 + a_3 + a_4), \quad \rho_1 = V_1,$$
$$V_2 \sim \mathrm{Beta}(a_2, a_3 + a_4), \quad \rho_2 = (1 - V_1)V_2,$$
$$V_3 \sim \mathrm{Beta}(a_3, a_4), \quad \rho_3 = (1 - V_1)(1 - V_2)V_3,$$

and so on, with

$$\rho_K = 1 - \sum_{k=1}^{K-1} \rho_k.$$

This is the *stick-breaking* representation.

2.8.2 Infinite Stick-Breaking

Taking $K \to \infty$, define

$$V_k \sim \mathrm{Beta}(a_k, b_k),$$

and construct weights:

$$\rho_k = \left(\prod_{j=1}^{k-1} (1 - V_j) \right) V_k.$$

2.9 GEM Distribution

A special case of stick-breaking is:

$$a_k = 1, \quad b_k = \alpha > 0.$$

Then

$$\rho = (\rho_1, \rho_2, \ldots) \sim \mathrm{GEM}(\alpha),$$

with

$$V_k \sim \mathrm{Beta}(1, \alpha), \quad \rho_k = \left(\prod_{j=1}^{k-1} (1 - V_j) \right) V_k.$$

This defines a random probability distribution over a countably infinite set.

2.10 Interpretation

Hierarchy of Distributions

- Beta: random distribution over $\{1, 2\}$
- Dirichlet: random distribution over $\{1, \ldots, K\}$
- GEM / stick-breaking: random distribution over $\{1, 2, \ldots\}$

Key Properties

- Infinitely many components
- Only finitely many used for finite data
- Number of active components grows with N

Conceptual Insight

- Parameters: infinite mixture components
- Effective parameters: number of occupied clusters

2.11 Readings: Foundations of the Dirichlet Process

Dirichlet Process as a Distribution over Distributions

A Dirichlet process defines a *random probability measure* P on a measurable space $(\mathcal{X}, \mathcal{B})$.

- P itself is the parameter of interest
- P takes values in the space of all probability measures

The key defining property is:
For any measurable partition $(B_1, \ldots, B_k)$,

$$\Big(P(B_1), \ldots, P(B_k)\Big) \sim \text{Dirichlet}\Big(a(B_1), \ldots, a(B_k)\Big),$$

where a is a finite base measure.

Interpretation of the Base Measure

Let

$$a(\mathcal{X}) = \alpha, \quad \beta(\cdot) = \frac{a(\cdot)}{\alpha}.$$

Then:

- β is the *base distribution* (mean measure)
- α controls concentration

Key intuition:

$$E[P(B)] = \beta(B)$$

Thus, the Dirichlet process is centered around β.

Discrete Nature of the Dirichlet Process

A fundamental property:

- With probability 1, P is *discrete*

  That is,

$$P = \sum_{n=1}^{\infty} p_n \delta_{Y_n}$$

  even if β is continuous.
  This is crucial:

- induces clustering
- explains repeated parameter values

Posterior Conjugacy

If

$$P \sim \mathrm{DP}(a), \quad X \mid P \sim P,$$

then the posterior is:

$$P \mid X \sim \mathrm{DP}(a + \delta_X).$$

For data $(X_1, \dots, X_n)$:

$$P \mid X_{1:n} \sim \mathrm{DP}\left(a + \sum_{i=1}^{n} \delta_{X_i}\right).$$

This is the infinite-dimensional analogue of Dirichlet conjugacy.

Constructive Representation (Stick-Breaking)

A constructive definition of P is:

$$P = \sum_{n=1}^{\infty} p_n \delta_{Y_n}$$

where:

- $Y_n \sim \beta$
- weights are given by stick-breaking:

$$p_1 = V_1, \quad p_n = V_n \prod_{j=1}^{n-1} (1 - V_j)$$

- $$V_n \sim \mathrm{Beta}(1, \alpha)$$

This gives an explicit generative construction of the random measure.

Distributional Fixed-Point Property

The Dirichlet process satisfies the distributional equation:

$$P = \theta\delta_Y + (1-\theta)P',$$

where:

- $\theta \sim \text{Beta}(1, \alpha)$
- $Y \sim \beta$
- $P' \stackrel{d}{=} P$ and independent

This recursive structure underlies stick-breaking.

Connection to Bayesian Nonparametrics

The Dirichlet process enables:

- modeling unknown distributions directly
- infinite mixture models
- automatic complexity control

Instead of fixing K, we obtain:

$$K = \infty \quad \text{but effective clusters } < \infty.$$

Practical Insight

- Only finitely many atoms receive non-negligible mass
- Sampling requires only a finite truncation
- Posterior updates remain tractable

This resolves the key issue:

$$\text{Unknown number of clusters} \Rightarrow \text{handled probabilistically.}$$

Distributions Hierarchy Interpretation

We can view the progression:

- Beta: random distribution over $\{1, 2\}$
- Dirichlet: random distribution over $\{1, 2, \ldots, K\}$
- GEM / stick-breaking: random distribution over $\{1, 2, \ldots\}$
- Dirichlet process: random distribution over Φ

Formally, the Dirichlet process admits the representation:

$$\rho = (\rho_1, \rho_2, \dots) \sim \mathrm{GEM}(\alpha), \quad \phi_k \overset{i.i.d.}{\sim} G_0,$$

$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\phi_k}.$$

Thus:

- ρ_k controls cluster weights
- ϕ_k are atom locations
- G is a discrete random measure

2.11.1 Dirichlet Process Mixture Model (DPMM)

Gaussian mixture case

- Weights:
$$\rho \sim \mathrm{GEM}(\alpha)$$
- Cluster parameters:
$$\mu_k \overset{i.i.d.}{\sim} \mathcal{N}(\mu_0, \Sigma_0)$$
- Random measure:
$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\mu_k} \equiv DP(\alpha, \mathcal{N}(\mu_0, \Sigma_0))$$

Latent structure:

$$z_n \overset{i.i.d.}{\sim} \mathrm{Categorical}(\rho), \quad \mu_n^* = \mu_{z_n}, \quad \mu_n^* \overset{i.i.d.}{\sim} G$$

Observations:

$$x_n \mid \mu_n^* \sim \mathcal{N}(\mu_n^*, \Sigma)$$

General formulation

More generally:

$$\rho \sim \mathrm{GEM}(\alpha), \quad \phi_k \overset{i.i.d.}{\sim} G_0$$

$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\phi_k} \equiv DP(\alpha, G_0)$$

$$z_n \sim \mathrm{Categorical}(\rho), \quad \theta_n = \phi_{z_n}, \quad \theta_n \overset{i.i.d.}{\sim} G$$

$$x_n \mid \theta_n \sim F(\theta_n)$$

Dirichlet Process Definition

Let $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$ be a measurable space.

Theorem 2.11.1. *(Dirichlet Process) A random probability measure F satisfies*

$$F \sim DP(\alpha, H)$$

if for any measurable partition $(B_1, \ldots, B_n)$,

$$(F(B_1), \ldots, F(B_n)) \sim \text{Dirichlet}(\alpha H(B_1), \ldots, \alpha H(B_n)).$$

Key implication:

$$p(F) \text{ is well-defined via Kolmogorov consistency.}$$

Properties of the Dirichlet Process

Moments

For any measurable A:

$$E[F(A)] = H(A), \quad \text{Var}(F(A)) = \frac{H(A)(1 - H(A))}{1 + \alpha}.$$

Self-similarity

For B such that $H(B) > 0$:

$$F_B \sim DP(\alpha, H_B),$$

where F_B is the restriction of F to B.

Conjugacy

If

$$y_i \overset{i.i.d.}{\sim} F, \quad F \sim DP(\alpha, H),$$

then

$$F \mid y \sim DP\left(\alpha + n, \ \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{y_i} + \frac{\alpha}{\alpha + n} H\right).$$

In particular:

$$E[F(A) \mid y] = \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{y_i}(A) + \frac{\alpha}{\alpha + n} H(A).$$

Richness

- Full support over distributions absolutely continuous w.r.t. H
- Can approximate arbitrary distributions

Stick-Breaking Theorem

Theorem 2.11.2. *(Sethuraman) Let*

$$\theta_k \overset{i.i.d.}{\sim} H, \quad V_k \overset{i.i.d.}{\sim} \mathrm{Beta}(1, \alpha),$$

and define

$$\pi_k = V_k \prod_{l=1}^{k-1} (1 - V_l).$$

Then

$$\sum_{k=1}^{\infty} \pi_k \delta_{\theta_k} \sim DP(\alpha, H).$$

Posterior Stick-Breaking Interpretation

Given

$$G \sim DP(\alpha, H), \quad \theta \sim G,$$

we obtain:

$$G \mid \theta \sim DP\left(\alpha + 1, \ \frac{\alpha H + \delta_\theta}{\alpha + 1}\right).$$

This implies a decomposition:

$$G = V\delta_\theta + (1 - V)G', \quad V \sim \mathrm{Beta}(1, \alpha),$$

where G' is the residual measure.

Recursive Property of the DP

For a partition $(\theta, A_1, \ldots, A_K)$:

$$(G(\theta), G(A_1), \ldots, G(A_K)) \sim \mathrm{Dirichlet}(1, \alpha H(A_1), \ldots, \alpha H(A_K)).$$

This implies:

$$G' \sim DP(\alpha, H),$$

showing the *self-similar recursive structure* of the Dirichlet process.

Key Insight

The Dirichlet process can be understood as:

- a distribution over discrete measures
- constructed via stick-breaking
- inducing clustering through shared atoms
- recursively decomposable via Beta splitting

This structure is what enables:

$$\text{infinite mixture models} \quad \Longleftrightarrow \quad \text{finite data-driven clustering.}$$

Stick-Breaking Expansion

We can recursively expand the Dirichlet process draw:

$$G \sim DP(\alpha, H)$$

$$G = V_1 \delta_{\theta_1^*} + (1 - V_1) G_1$$

$$G = V_1 \delta_{\theta_1^*} + (1 - V_1)\Big(V_2 \delta_{\theta_2^*} + (1 - V_2) G_2\Big)$$

Continuing:

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*},$$

where

$$\pi_k = V_k \prod_{l=1}^{k-1} (1 - V_l), \quad V_k \sim \text{Beta}(1, \alpha), \quad \theta_k^* \sim H.$$

Interpretation

- Draws from a DP are *discrete* measures
- Mass is allocated via stick-breaking
- Atoms θ_k^* are drawn from the base distribution H

Marginalizing the Weights

Consider a simple case:

$$\rho_1 \sim \text{Beta}(a_1, a_2), \quad z_n \sim \text{Categorical}(\rho_1, \rho_2).$$

Integrating out ρ_1, we obtain:

$$p(z_n = 1 \mid z_{1:n-1}) = \frac{a_{1,n}}{a_{1,n} + a_{2,n}},$$

where

$$a_{1,n} = a_1 + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = 1\}, \quad a_{2,n} = a_2 + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = 2\}.$$

2.11.2 Pólya Urn Interpretation

This predictive rule corresponds to a Pólya urn:

- Choose a ball with probability proportional to its count
- Replace it and add another ball of the same color

Then:

$$\lim_{n \to \infty} \frac{\#\ \text{orange}}{\#\ \text{total}} \overset{d}{=} \text{Beta}(a_{\text{orange}}, a_{\text{green}}).$$

Multivariate Extension

For

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}), \quad z_n \sim \text{Categorical}(\rho_{1:K}),$$

we obtain:

$$p(z_n = k \mid z_{1:n-1}) = \frac{a_{k,n}}{\sum_{j=1}^{K} a_{j,n}},$$

where

$$a_{k,n} = a_k + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = k\}.$$

This corresponds to a *multivariate Pólya urn*.

2.11.3 Blackwell–MacQueen Pólya Urn

We now consider the infinite limit.

Let:

$$G \sim DP(\alpha, H), \quad \theta_i \mid G \sim G.$$

Then:

First sample

$$\theta_1 \sim H, \quad G \mid \theta_1 \sim DP\left(\alpha + 1, \frac{\alpha H + \delta_{\theta_1}}{\alpha + 1}\right).$$

Second sample

$$\theta_2 \mid \theta_1 \sim \frac{\alpha H + \delta_{\theta_1}}{\alpha + 1}.$$

General step

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1}.$$

Urn Interpretation

- With probability proportional to α, draw a new value from H
- With probability proportional to existing counts, reuse a previous value

Equivalently:

- Draw a ball proportional to its mass
- If it is *new*, create a new color
- Otherwise, reinforce the existing color

Key Insight

This produces a sequence:

$$\theta_1, \theta_2, \ldots$$

with:

- clustering (shared values)
- increasing number of clusters
- probability of new cluster controlled by α

2.12 Chinese Restaurant Process (CRP)

Construction

The Chinese Restaurant Process provides an equivalent formulation of the Dirichlet process clustering mechanism.

- Each observation (customer) arrives sequentially.
- Customer n:
  - sits at an existing table k with probability proportional to the number of customers already there,
  - starts a new table with probability proportional to α.

Formally:

$$p(z_n = k \mid z_{1:n-1}) = \begin{cases} \frac{n_k}{n-1+\alpha}, & \text{existing table } k, \\ \frac{\alpha}{n-1+\alpha}, & \text{new table.} \end{cases}$$

Partition Interpretation

Let Π_N denote the partition of $\{1, \ldots, N\}$ induced by assignments $z_1, \ldots, z_N$.
Example:

$$z_1 = z_2 = z_7 = z_8 = 1, \quad z_3 = z_5 = z_6 = 2, \quad z_4 = 3$$

$$\Rightarrow \Pi_8 = \{\{1, 2, 7, 8\}, \{3, 5, 6\}, \{4\}\}.$$

A partition consists of mutually exclusive and exhaustive subsets of $\{1, \ldots, N\}$.

Probability of a Seating Arrangement

Let K_N be the number of tables and let $C \in \Pi_N$ denote a cluster.
Then:

$$\mathbb{P}(\Pi_N = \pi_N) = \frac{\alpha^{K_N} \prod_{C \in \Pi_N} (|C| - 1)!}{\alpha(\alpha + 1) \cdots (\alpha + N - 1)}.$$

Exchangeability

The probability of a partition does not depend on the order of customers:

$$\mathbb{P}(\Pi_N = \pi_N) \text{ is invariant under permutations.}$$

Thus, the CRP is *exchangeable*.

Conditional Assignment Rule

Given $\Pi_{N,-n}$ (partition without customer n):

$$p(\Pi_N \mid \Pi_{N,-n}) = \begin{cases} \frac{|C|}{\alpha+N-1}, & \text{if } n \text{ joins cluster } C, \\ \frac{\alpha}{\alpha+N-1}, & \text{if } n \text{ creates new cluster.} \end{cases}$$

Representations of the Dirichlet Process

The Dirichlet process admits several equivalent representations:

- *Posterior form:*

$$G \sim DP(\alpha, H), \quad \theta \mid G \sim G$$

$$\iff \theta \sim H, \quad G \mid \theta \sim DP\left(\alpha + 1, \frac{\alpha H + \delta_\theta}{\alpha + 1}\right)$$

- *Pólya urn:*

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1}$$

- *CRP:*

$$p(z_n = k \mid z_{1:n-1}) = \begin{cases} \frac{n_k}{n-1+\alpha}, \\ \frac{\alpha}{n-1+\alpha} \end{cases}$$

- *Stick-breaking:*

$$\pi_k = \beta_k \prod_{i=1}^{k-1} (1 - \beta_i), \quad \beta_k \sim \text{Beta}(1, \alpha)$$

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*}$$

Exchangeability and Joint Distribution

From the predictive rule:

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1},$$

the joint distribution is:

$$p(\theta_1, \ldots, \theta_n) = \frac{\alpha^K \prod_{k=1}^K h(\theta_k^*)(m_k - 1)!}{\prod_{i=1}^n (\alpha + i - 1)},$$

where:

- θ_k^* are distinct values,
- m_k are their multiplicities,
- $h(\theta)$ is the density under H.

This distribution is exchangeable.
By De Finetti's theorem, this implies the existence of a random measure G such that

$$\theta_1, \theta_2, \ldots \overset{iid}{\sim} G, \quad G \sim DP(\alpha, H).$$

2.13 Number of Clusters

Let K_n denote the number of clusters among n observations.

2.13.1 Expected Value

Define indicator:

$$I_i = \mathbf{1}\{\text{new cluster at step } i\}.$$

Then:

$$\mathbb{E}[K_n] = \sum_{i=1}^n \mathbb{E}[I_i] = \sum_{i=1}^n \frac{\alpha}{\alpha + i - 1}.$$

Thus:

$$\mathbb{E}[K_n] = O(\alpha \log n).$$

2.13.2 Variance

$$\mathrm{Var}(K_n) = \sum_{i=1}^n \frac{\alpha(i-1)}{(\alpha + i - 1)^2}.$$

2.13.3 Distribution of K_n

The distribution satisfies:

$$\mathbb{P}(K_n = k) = \frac{\alpha}{\alpha + n - 1}\mathbb{P}(K_{n-1} = k - 1) + \frac{n-1}{\alpha + n - 1}\mathbb{P}(K_{n-1} = k).$$

Boundary conditions:

$$\mathbb{P}(K_1 = 1) = 1.$$

Closed form:

$$\mathbb{P}(K_n = k) = C_n(k)\, \alpha^k \frac{\Gamma(\alpha)}{\Gamma(\alpha + n)},$$

where $C_n(k)$ are unsigned Stirling numbers of the first kind.

2.13.4 Interpretation

- Rich-get-richer effect: larger clusters attract more points
- Few large clusters, many small ones
- Number of clusters grows sublinearly in n

Chapter 3

Sampling Techniques

3.1 Sampling Techniques for DP Mixture Models

Gaussian Likelihood Setup

Assume:

$$y \sim \mathcal{N}(\mu, \sigma^2), \quad p(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y-\mu)^2\right).$$

For i.i.d. data $y = (y_1, \ldots, y_n)$:

$$L(\mu, \sigma^2) \propto \left(\frac{1}{\sigma^2}\right)^{n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_i (y_i - \mu)^2\right).$$

We focus on:

$$p(\mu, \sigma^2 \mid y) = p(\mu \mid \sigma^2, y)\, p(\sigma^2 \mid y).$$

Semi-Conjugate Prior

Assume independent priors:

$$\pi(\mu, \sigma^2) = \pi(\mu)\pi(\sigma^2),$$

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad \sigma^2 \sim IG\left(\frac{\nu_0}{2}, \frac{\delta_0}{2}\right).$$

Then:

$$\mu \mid \sigma^2, y \sim \mathcal{N}(\mu_n, \tau_n^2),$$

$$\mu_n = \frac{\mu_0/\sigma_0^2 + \bar{y}\, n/\sigma^2}{1/\sigma_0^2 + n/\sigma^2}, \quad \tau_n^2 = \frac{1}{1/\sigma_0^2 + n/\sigma^2}.$$

$$\sigma^2 \mid y \text{ not in closed form}, \quad \sigma^2 \mid y, \mu \sim IG.$$

3.1.1 Finite Mixture Models

A flexible density:

$$f(y) = \sum_{k=1}^{K} \tau_k f(y \mid \theta_k),$$

where:

$$\tau_k \in (0,1), \quad \sum_k \tau_k = 1.$$

3.1.2 Bayesian Mixture Model

$$y_i \mid \tau, \theta \sim \sum_{k=1}^{K} \tau_k f(y_i \mid \theta_k),$$

$$\tau \sim \mathrm{Dir}(\alpha, \dots, \alpha), \quad \theta_k \sim \pi_k(\theta_k), \quad K \sim \pi(K).$$

3.1.3 Gibbs Sampling (Finite Case)

For fixed K:

- Assignment:
$$P(z_i^{(t)} = k \mid \cdot) \propto \tau_k^{(t-1)} f(y_i \mid \theta_k^{(t-1)}).$$
- Weights:
$$\tau \mid z \sim \mathrm{Dir}(\alpha_1 + \hat{n}_1, \dots, \alpha_K + \hat{n}_K),$$
where $\hat{n}_k = \sum_i \mathbf{1}(z_i = k)$.
- Parameters:
$$p(\theta_k \mid \cdot) \propto \pi_k(\theta_k) \prod_{i:z_i=k} f(y_i \mid \theta_k).$$

3.2 CRP Mixture Model

Generative Model

$$\Pi_N \sim CRP(N, \alpha),$$

$$\forall C \in \Pi_N, \quad \phi_C \overset{i.i.d.}{\sim} G_0,$$

$$x_n \mid \phi_C \sim F(\phi_C), \quad n \in C.$$

Inference Goal

$$p(\Pi_N \mid x_{1:N}).$$

Gibbs Sampler (Collapsed)

$$p(\Pi_N \mid \Pi_{N,-n}, x) = \begin{cases} \frac{|C|}{\alpha+N-1}\, p(x_{C\cup\{n\}} \mid x_C), & \text{join } C, \\ \frac{\alpha}{\alpha+N-1}\, p(x_n), & \text{new cluster.} \end{cases}$$

For Gaussian case:

$$p(x_{C\cup\{n\}} \mid x_C) = \mathcal{N}(\tilde{m}, \tilde{\Sigma} + \Sigma),$$

$$\tilde{\Sigma}^{-1} = \Sigma_0^{-1} + |C|\Sigma^{-1}, \quad \tilde{m} = \tilde{\Sigma}\left(\Sigma^{-1}\sum_{m\in C} x_m + \Sigma_0^{-1}\mu_0\right).$$

3.3 DP Mixture Model (DPM)

$$y_i \sim \int p(y_i \mid \theta) G(d\theta), \quad G \sim DP(M, G_0).$$

Example:

$$y_i \sim \int \mathcal{N}(y_i \mid \mu, \sigma^2) G(d\mu).$$

3.4 Alternative Representations

3.4.1 Stick-Breaking Form

$$y_i \mid (w_h, \tilde{\theta}_h) \sim \sum_{h=1}^{\infty} w_h p(y_i \mid \tilde{\theta}_h),$$

$$\tilde{\theta}_h \sim G_0, \quad w_h = v_h \prod_{k<h}(1 - v_k), \quad v_h \sim \text{Beta}(1, M).$$

3.4.2 Clustering Representation

$$y_i \mid \theta_i \sim p(y_i \mid \theta_i), \quad \theta_i \mid G \sim G, \quad G \sim DP(M, G_0).$$

Integrating out G:

$$(\theta_1, \ldots, \theta_n) \sim \text{CRP-induced distribution.}$$

Predictive:

$$p(\theta_{n+1} \mid \theta_{1:n}) \propto \sum_{j=1}^{k_n} n_j \delta_{\theta_j^*} + MG_0.$$

3.5 Collapsed Gibbs Sampling

3.5.1 Conditional for Parameters

$$\theta_i \mid \theta_{-i}, y \propto \sum_{j=1}^{k^-} n_j^- p(y_i \mid \theta_j^*)\delta_{\theta_j^*} + Mp(y_i \mid \theta_i)G_0(\theta_i).$$

Posterior:

$$p(\theta_i \mid y_i, G_0) = \frac{p(y_i \mid \theta_i)G_0(\theta_i)}{\int p(y_i \mid \theta)G_0(d\theta)}.$$

3.5.2 Cluster Assignment Update

$$p(s_i = j \mid s^-, y) \propto \begin{cases} n_j^- \int p(y_i \mid \theta_j^*)dp(\theta_j^* \mid y_j^-), & j \leq k^-, \\ M \int p(y_i \mid \theta)dG_0(\theta), & j = k^- + 1. \end{cases}$$

3.5.3 Cluster Parameter Update

$$p(\theta_j^* \mid s, y) \propto G_0(\theta_j^*) \prod_{i:s_i=j} p(y_i \mid \theta_j^*).$$

3.6 Gaussian DPM Example

$$p(y_i \mid \theta_i) = \mathcal{N}(\theta_i, \sigma^2), \quad G_0 = \mathcal{N}(m, B).$$

$$\int p(y_i \mid \theta)dG_0(\theta) = \mathcal{N}(y_i \mid m, B + \sigma^2).$$

Cluster predictive:

$$\int p(y_i \mid \theta_j^*)dp(\theta_j^* \mid y_j^-) = \mathcal{N}(y_i \mid m_j^-, V_j^- + \sigma^2),$$

$$V_j^- = \left(\frac{1}{B} + \frac{n_j^-}{\sigma^2}\right)^{-1}, \quad m_j^- = V_j^- \left(\frac{m}{B} + \frac{1}{\sigma^2}\sum_{h \in S_j^-} y_h\right).$$

Posterior:

$$p(\theta_j^* \mid y) = \mathcal{N}(m_j, V_j).$$